

a vitrectomy reduces the postoperative inflammation and breakdown of the blood–ocular barrier.⁴ These two factors may work in the current case, at least in part.

Some controversy remains concerning the cause of retinal detachment in MGS. A few reported cases have clearly identifiable retinal tears, causing a rhegmatogenous retinal detachment.⁵ Even in cases in which a retinal break is unclear, there is some evidence that a small retinal break may be a source of subretinal fluid from the vitreous cavity.⁶ Proposed sources include the transudation from the abnormal retinal vessels, liquid vitreous passing through an optic disk cup into the subretinal space, or cerebrospinal fluid communicating between the subarachnoid space and subretinal space. In the current study, neither a retinal break nor a tear was found preoperatively or intraoperatively, and postoperative fluorescein angiography indicated no vascular leakage from the subretinal vessels. The retina remained successfully reattached more than 1 year after a single operation, whereas in most previous cases, multiple surgeries are necessary.² This may be because sufficient removal of the preretinal posterior hyaloid could not be achieved in the previous cases. Because of these findings, the tractional force by preretinal posterior hyaloid may be an important factor in the occurrence of retinal detachment in MGS. This notion is compatible with the one that nonrhegmatogenous retinal detachment in MGS is probably the result of the continuous traction exerted by the gradually increasing axial retro-displacement of the optic nerve by Manschot.⁷ Therefore, sufficient removal of the posterior hyaloid and the prevention of postoperative inflammation by a TA-assisted pars plana vitrectomy is considered to be an effective treatment for nonrhegmatogenous retinal detachment in MGS.

The exact pathogenesis of MGS is still unclear, but a mesodermal disorder is speculated to be one possible cause because the peripapillary fibrous tissue has been reported to consist of thick collagen fibers with glial cells and retinal pigment epithelial cells.⁷ Although no apparent pigment epithelial cells were found in the current specimen, the histologic features, including glial cells associated with abundant collagen fibers and vessels, were consistent with the findings of previous reports.

In conclusion, a TA-assisted pars plana vitrectomy is considered to be a potentially useful treatment for retinal detachment in MGS.

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DISAPPEARANCE OF RETINAL MYELINATED NERVE FIBERS AFTER PLAQUE RADIOTHERAPY FOR CHOROIDAL MELANOMA

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A patch of retinal myelinated nerve fibers slowly disappeared after development of radiation retinopathy and papillopathy after plaque radiotherapy for choroidal melanoma.

Case Report

In 1990, macular choroidal melanoma was diagnosed in the right eye of a 69-year-old woman. She also had myelinated retinal nerve fibers superonasal to the right optic disk (Figure 1). The melanoma was treated with an iodine 125 plaque, delivering 9,000 cGy to the tumor apex and 43,200 cGy to the tumor base over 146 hours. In 1991, the tumor was regressed, the myelinated retinal fibers were unchanged, and one nerve fiber layer infarction was present in the macular region. In 1992, radiation retinopathy with intraretinal hemorrhages and occlusion of inferotemporal retinal vessels were present. The myelinated fibers showed small areas of fading (Figure 2). In 1994, the visual acuity was hand motions, and radiation papillopathy with a swollen optic disk and peripapillary hemorrhages were evident. The myelinated retinal fibers were less apparent. In 1996, the optic disk was pale, inferior retinal vessels were atrophic, and the myelinated nerve fibers had disappeared. In 2002, the findings were stable (Figure 3).

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Fig. 1. At presentation (1990), a patch of retinal myelinated nerve fibers is visible superonasal to the optic disk of the right eye. A pigmented choroidal melanoma is also present in the macular area.

Discussion

Although retinal myelinated nerve fibers are generally stable, few reports have documented their disappearance in conditions such as glaucoma,¹ optic neuritis,² ischemic optic neuropathy,³ central retinal artery occlusion,⁴ and Behçet disease.⁵ The common mechanism in these conditions appears to be optic nerve and retinal nerve fiber layer atrophy.

In this report, we show photographic documentation of retinal myelin loss after plaque radiotherapy for choroidal melanoma. Loss of myelin was first noted 2 years after radiotherapy and 1 year after onset of radiation retinopathy, and it continued after onset of

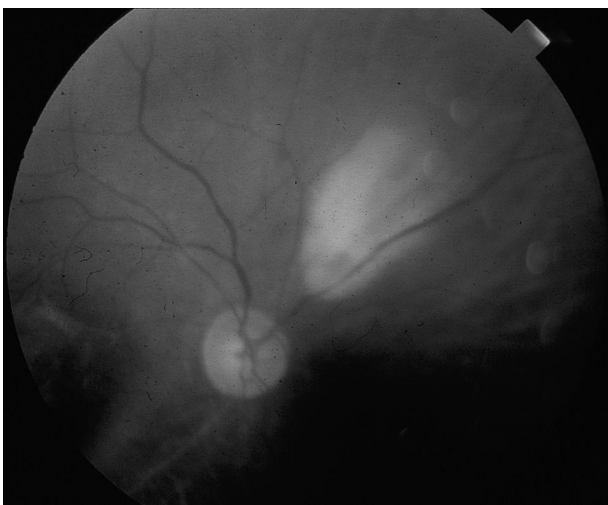


Fig. 2. Two years after plaque radiotherapy (1992), small areas of involution are noted in the myelin. The inferotemporal retinal vessels are occluded, and the choroidal melanoma has shown regression.

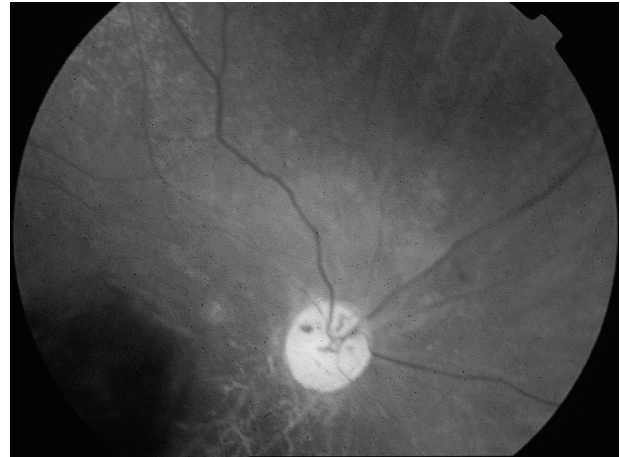


Fig. 3. Twelve years after plaque radiotherapy (2002), the myelinated nerve fibers have disappeared, leaving atrophic retina. The inferior retinal vessels are occluded, and the optic disk is pale with small hemorrhages on its surface.

radiation papillopathy and optic atrophy. Although loss of oligodendrocytes is a known complication of radiation of the central nervous system,⁶ the time course in our case suggests that loss of retinal myelin occurred as a result of atrophy of retinal nerve fibers after radiation retinopathy and papillopathy.

Key words: demyelination, eye, medullated nerve fibers, myelinated nerve fibers, optic atrophy, radiation papillopathy, radiation retinopathy, radiotherapy, retina.

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